Storytelling as Inverse Inverse Planning

Chandra, Li, Tenenbaum & Ragan-Kelley (2023), read with Borges, *The Garden of Forking Paths* (1941)

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Notes and slides

In one sentence

A storyteller plans for the audience's beliefs, not for the characters' goals: it chooses actions so that the audience's inferences, modelled as Bayesian theory of mind, follow a designed path.

Motivation

Viewers see agents with goals in simple moving shapes (Heider & Simmel, 1944). A rational planner reaches its goals, but an efficient path leaves no doubt, so it has no suspense and no irony. Kukkonen (2014): a story is a "probability design" of the reader's beliefs.

Three levels

Planning	goal → actions (act)
Inverse planning	actions → $\mathrm{P}(\text{goal} \mid \text{actions})$ (ToM)
Inverse inverse	objective → actions, via an audience model

Audience model (Ullman et al., 2009)

Grid world: a robot that can push objects and a weak cheese (its moves succeed 60% of the time), goal tiles pink and green. The robot's reward includes $\rho \cdot r_{\text{cheese}}$, the cheese's reward weighted by ρ: $\rho > 0$ helps, $\rho < 0$ hinders, $\rho = 0$ indifferent. The robot reasons about a cheese that reasons about the robot. An observer infers the hypothesis H (both goals G and ρ) from each action a in state s:

$$\mathrm{P}(H \mid s \to a) \propto \mathrm{P}(s \to a \mid H)\,\mathrm{P}(H)$$

Optimizing a script

A script σ is a setting plus a sequence of moves and their outcomes, all optimized; $\sigma_{1:t}$ is the part seen so far. Beam search (keep the best few partial scripts) maximizes, for example,

$$f_{\text{help}}(\sigma) = \textstyle\sum_t \mathrm{P}(\rho > 0 \mid \sigma_{1:t}).$$

The objective fixes no goals or positions. The optimizer finds a scene that shows helping.

Limits and critique

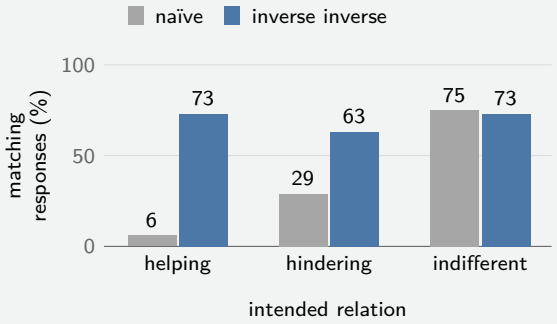
- **Paper**: scales badly; too slow for real time; emotion is future work.
- **Evidence** (critique, not from the paper): weak baseline; devices are one-seed demos, untested on humans. The method *can* make readable scenes; it does not show that people plan this way.
- **Borges** (interpretation): the Chief must already hold the hypothesis "a killing encodes a name"; one lossy report, not a full trajectory.
- **Ethics** (extension, not from the paper): the same machinery can deceive.

For discussion

1. What data would show that *people* tell stories this way?
2. Which objective f makes Yu Tsun's murder score highest?
3. Is the Chief a Bayesian inverse planner? With what prior?
4. Could an LLM be the audience model? How to test it?

Results

Exp. 1: 98 participants, 120 animations, 15 per person. Indifference is easy (characters rarely interact); not all scenes allow helping.



Mean of 20 animations per bar. Helping, hindering: $p < 0.01$; indifferent: n.s.

Exp. 2 ("inverse inverse physics"): a hopping robot mimes pulling a heavy box. Controls: chance on same-weight pairs, 97% on honest heavy vs. light. The mime is judged heavier than an honest light box (**95.7%**) and than an honest *heavy* box (**68.6%**). 35 participants in the text; Fig. 4b says N = 70.

Storytelling devices as objectives

- **Plot twist**: $\mathrm{P}(\rho > 0)$ in the first half, $\mathrm{P}(\rho < 0)$ in the second.
- **Dramatic irony**: the audience knows the cheese's goal; the robot does not and hinders by mistake.
- **Flashback**: append a known event to σ; find a past that makes it look helpful.
- **Narrative arc**: fortune rises, falls, rises; a KL penalty on belief changes keeps the goals stable; a table can fall "from the sky" that no character expects (deus ex machina).

The Garden of Forking Paths (an interpretation)

Yu Tsun, a German spy in England, must tell his Chief in Berlin the name of a town. He kills a stranger named Albert; the newspapers report it; the Chief understands and the town of Albert is bombed.

- **Reading 1**: the murder is inverse inverse planning, with storyteller Yu Tsun, audience the Chief, observations = newspapers. The act is terrible in the world but is the only one that creates the belief: an extreme mime.
- **Reading 2**: Borges shapes the reader's belief the same way: Yu Tsun calls himself Albert's friend, then shoots (twist).
- **Reading 3**: Ts'ui Pen's novel takes *every* branch; a storyteller keeps *one* script and lets the audience infer the meaning: show, don't tell.

Seminar: Bayesian ToM 03.12. ▪ LLM planning 17.12. ▪ `memo` 14.01. ▪ false belief 29.10. ▪ Clever Hans 05.11. ▪ awareness 07.01. ▪ epistemic logic 04.02.

Chandra, K., Li, T.-M., Tenenbaum, J. B., & Ragan-Kelley, J. (2023). Storytelling as inverse inverse planning. *Proc. CogSci* 45 (CC BY 4.0); code and videos: `https://people.csail.mit.edu/kach/a2i2p/`. Borges, J. L. (1941). El jardín de senderos que se bifurcan. Heider, F., & Simmel, M. (1944). An experimental study of apparent behavior. *Am. J. Psychol.* 57. Ullman, T. et al. (2009). Help or hinder. *NeurIPS*. Kukkonen, K. (2014). Bayesian narrative. *Anglia* 132.